DSA06 - Data Handling and Visualization

Activity: Identify Skewness using Violin Plots | Slot D

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Part A (18 marks)

Chart 1: Resting heart rate (beats per min)

Violin	Skew (circle one)	Mean vs Median (circle)
Club B	Right skew	Mean > Median
Club D	Symmetric	Mean ≈ Median
Club C	Left skew	Mean < Median

Chart 2: Order value (Rs hundred)

Violin	Skew (circle one)	Mean vs Median (circle)
Store 3	Symmetric	Mean ≈ Median
Store 4	Left skew	Mean < Median
Store 5	Right skew	Mean > Median

Chart 3: Delivery time (minutes)

Violin	Skew (circle one)	Mean vs Median (circle)
Zone S	Symmetric	Mean ≈ Median
Zone E	Right skew	Mean > Median
Zone N	Left skew	Mean < Median

Chart 4: Fuel efficiency (km per litre)

Violin	Skew (circle one)	Mean vs Median (circle)
Brand Z	Symmetric	Mean ≈ Median
Brand X	Left skew	Mean < Median
Brand Y	Right skew	Mean > Median

Part B (2 marks)

B1. In Chart 4, violin "Brand X" is not symmetric. Towards which side is its tail, and which is the better measure of a typical fuel efficiency here: mean or median? Give one reason.

Ans: The tail of Brand X points towards the lower side (left / downwards, i.e. lower fuel efficiency values), so it is left-skewed. The median is the better measure of typical fuel efficiency because it is not affected by the long tail of low values, whereas the mean gets pulled down by it.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown.

```
ggplot(df, aes(x = group, y = value, fill = group)) + geom_violin() +
stat_summary(fun = median, geom = "point")
```